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UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA
Faculty of Engineering, Built Environment and Information Technology

ELI 220

Practical Assignment 1

Group 16

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March 4, 2026

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1 Introduction

The Fourier series was used to approximate four different functions that being, $x(t) = sin(2\pi t)$, a square wave with 50% duty cycle, square wave with 10% duty cycle and saw-tooth with function $x(t) = t$. Further it is needed to be determined how many components of the Fourier series of the above mentioned functions are needed to create an accurate approximation of said functions.

2 Assignment and Results

2.1 Question 1

$$x(t) = sin(2\pi t)$$
$$T_0 = 1$$
$$\omega_0 = 2\pi$$

$$f(-x) = -f(x) \text{ function is odd.}$$

$$a_0 = 0$$
$$a_n = 0$$

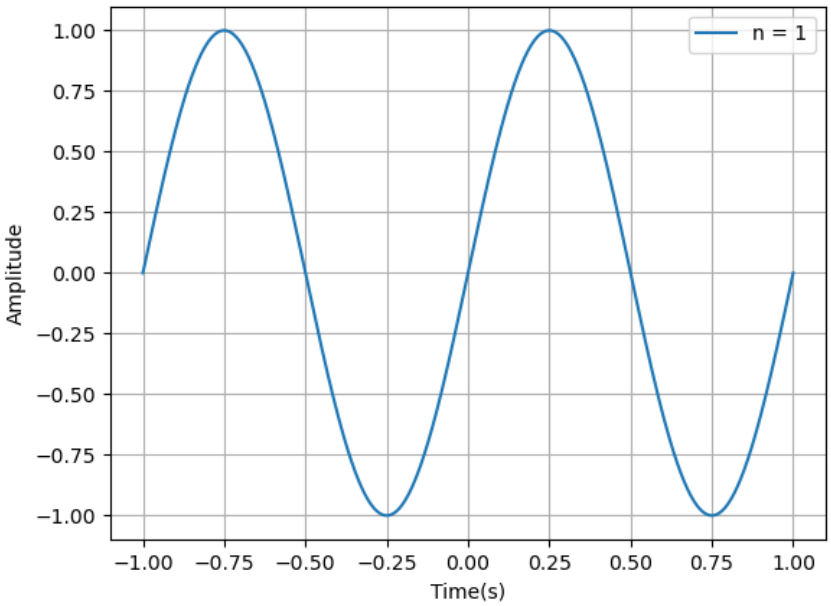
$$\begin{aligned}
b_n &= \frac{2}{T_0}(\int_{T_0} x(t)sin(2\pi nt)\ dt) \\
&= \frac{2}{1}(\int_0^1 (cos(2\pi t - 2\pi nt)\ cos(2\pi t + 2\pi nt)\ dt) \\
&= \frac{1}{2\pi - 2\pi n} sin(2\pi - 2\pi n) - \frac{1}{2\pi + 2\pi n} sin(2\pi + 2\pi n) \\
&= \frac{\frac{sin(2\pi(n-1))}{n-1} - \frac{sin(2\pi(n+1))}{n+1}}{4\pi} \\
&= \frac{0-0}{4\pi} \ = 0
\end{aligned}$$

$$b_n = 0 \text{ for } n > 1$$
$$b_1 = sin(2\pi t)$$

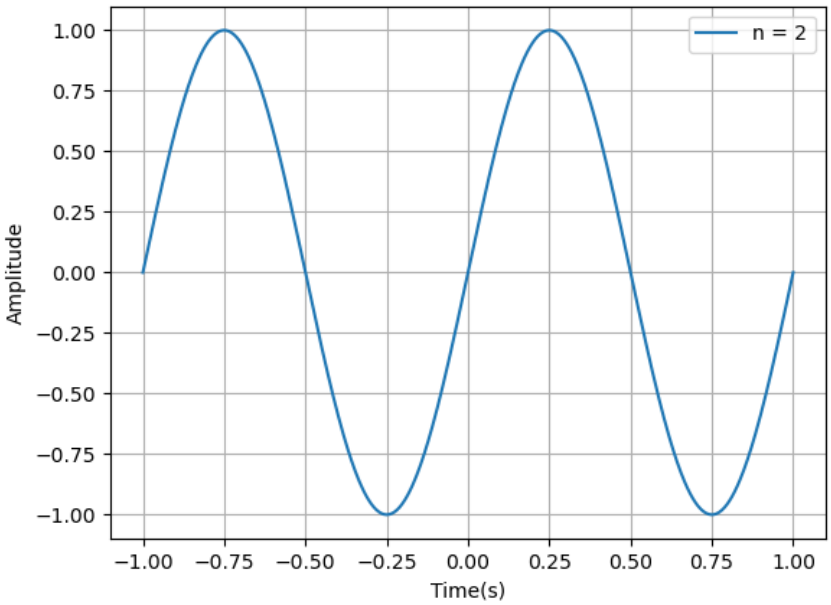
thus,

$$x(t) = \sum_{n=0}^{\infty} sin(2\pi t)$$

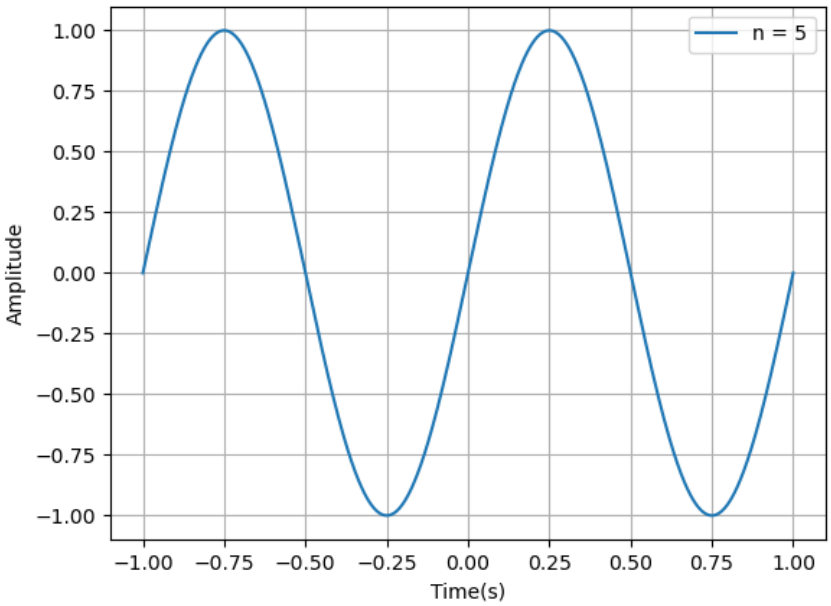
Question 1.1



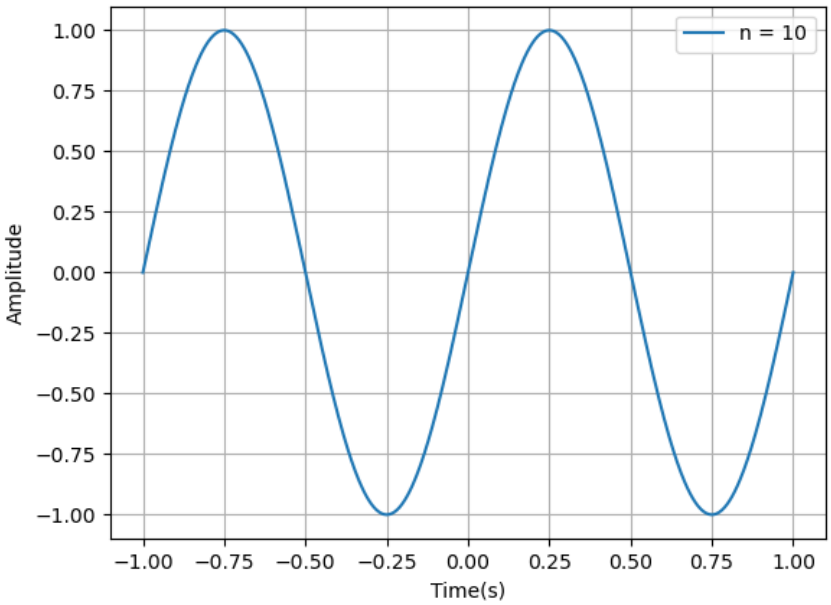
Question 1.2



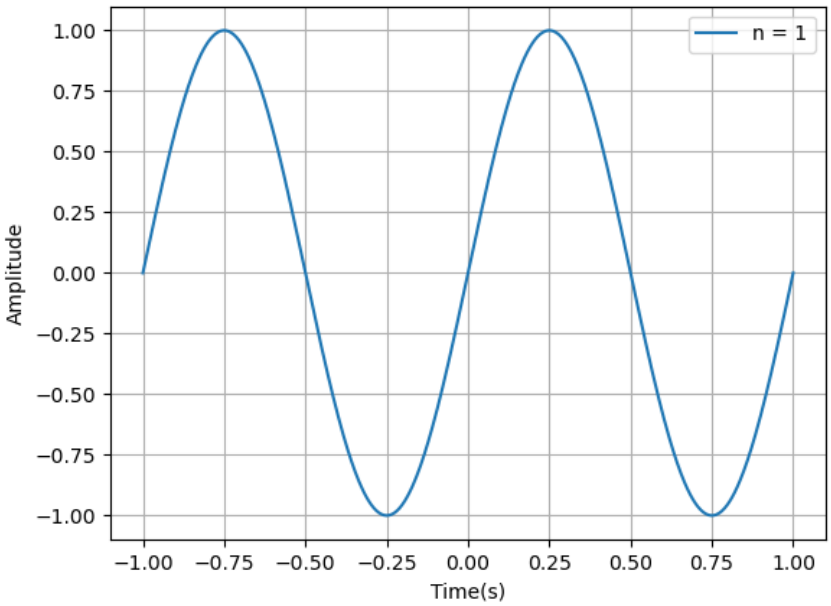
Question 1.3



Question 1.4



Question 1.5



Only one component is needed to determine the Fourier series of $x(t) = sin(2\pi t)$ as shown in the above calculations $b_n = 0$ for all $n > 1$ thus, only one component is needed to represent the Fourier series of $x(t) = sin(2\pi t)$

2.2 Question 2

$$T = 1$$
$$\omega_0 = -2\pi$$
$$a_0 = 0$$
$$a_n = 0 \text{ (odd)}$$

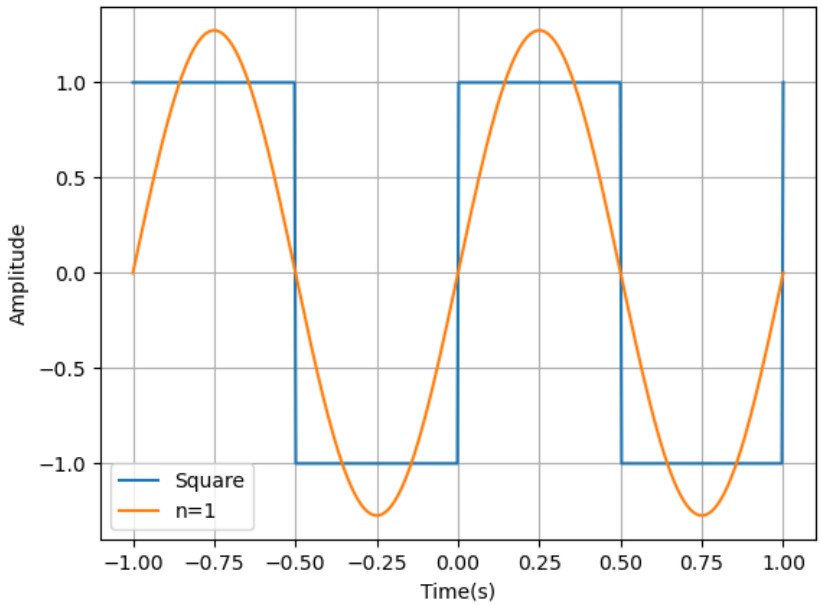
$$
\begin{aligned}
b_n &= \frac{2}{1}(\int_{0}^{0.5} sin(2\pi nt)\ dt - \int_{0.5}^{1} sin(2\pi nt)\ dt) \\
&= 2(\frac{-cos(2\pi nt)}{2\pi nt})|_0^{0.5} + 2(\frac{cos(2\pi nt)}{2\pi nt})|_{0.5}^{1} \\
&= (\frac{-cos(\pi n)}{\pi n} + \frac{1}{\pi n}) + (\frac{-cos(2\pi n)}{\pi n} - \frac{cos(\pi n)}{\pi n}) \\
&= \frac{1 - 2cos(\pi n) - cos(2\pi n)}{\pi n} \\
&= \frac{-2(cos(\pi n) + 1 + 1)}{\pi n} \text{ (see reason bellow)} \\
&= \frac{-2(cos(\pi n) + 2)}{\pi n} \\
&= \frac{-2}{\pi n} \times (cos(\pi n) - 1) \\
&= \frac{-2}{\pi n} \times ((-1)^n - 1)
\end{aligned}
$$

$cos(\pi n)$ goes to -1 for all odd n and +1 for all even n.
$cos(2\pi n)$ goes to +1 for all even n and -1 for all odd n.

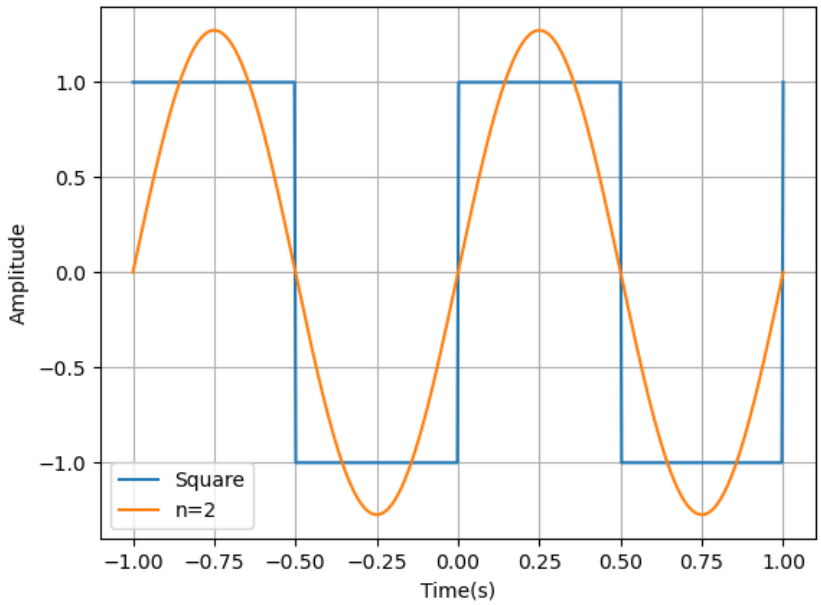
Thus, $cos(\pi n)$ can be taken out and later replaced by $(-1)^n$ as it can be either +1 or -1 depending on n as n will make both have the same value.

$$
x(t) = \sum_{n=1}^{\infty}(\frac{-2((-1)^n - 1)}{\pi n} cos(2\pi nt + \frac{\pi}{2}))
$$

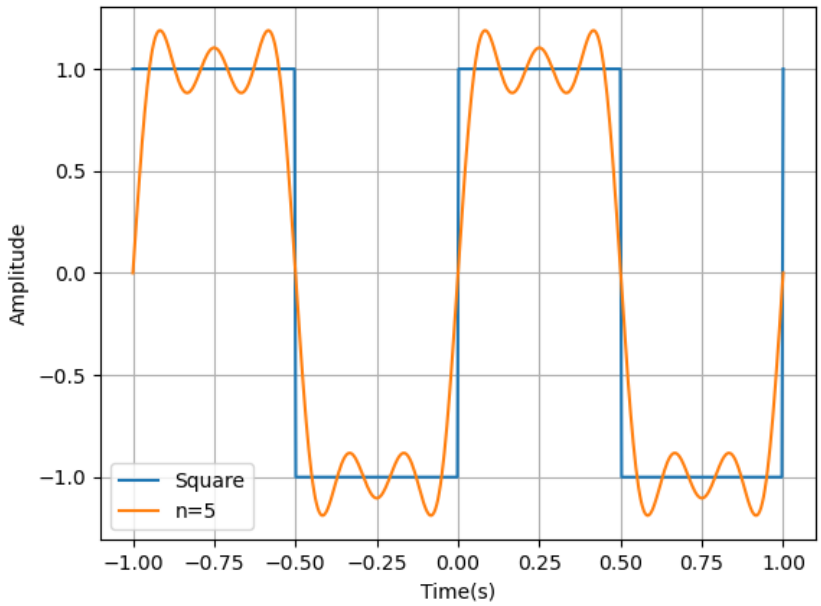
Question 2.1



Question 2.2



Question 2.3



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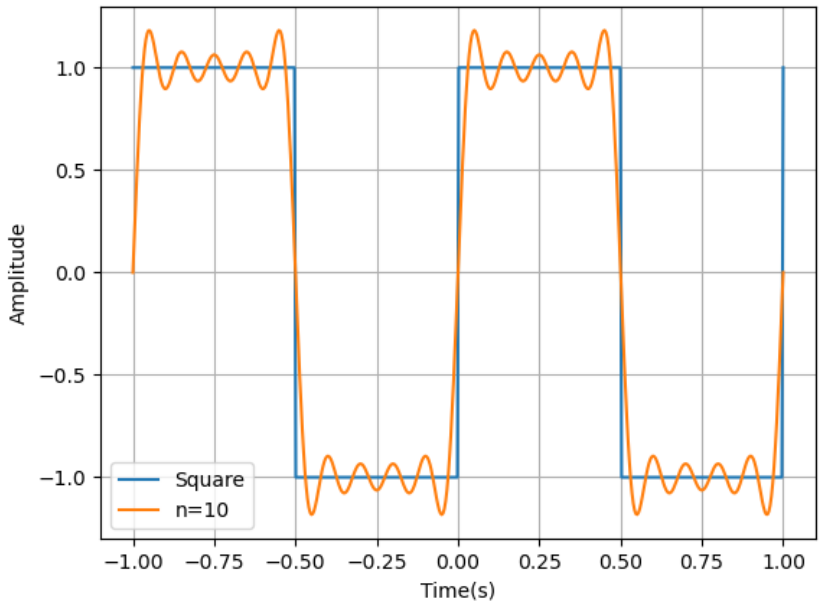

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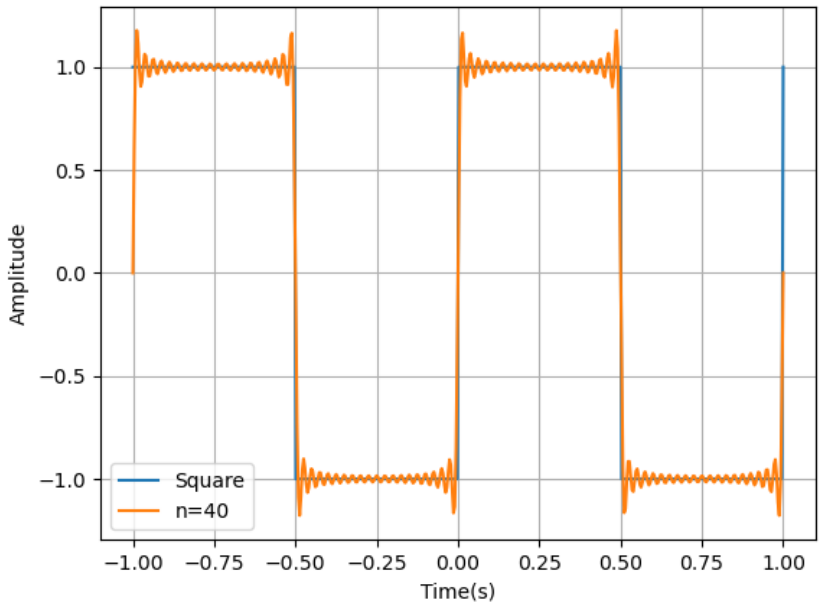


$$x(t) = \sum_{n=1}^{\infty}(\frac{-2((-1)^n - 1)}{\pi n} cos(2\pi nt + \frac{\pi}{2}))$$

With n form 1 to 40 in the above Fourier series we will get a good approximation of a square wave with 50% duty.

Thus we can determine forty components are needed to form a good graphical representation of a square wave with 50% duty cycle.

2.3 Question 3

$$T = 1$$
$$\omega_0 = 2\pi$$

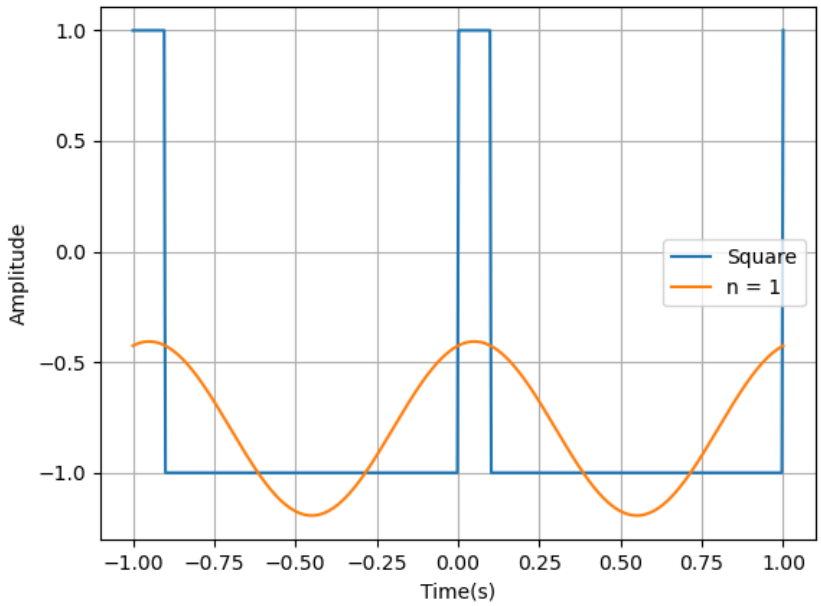
$$\begin{aligned} a_0 &= \int_0^{0.1} 1 \ dt + \int_{0.1}^{1} -1 \ dt \\ &= (t)|_0^{0.1} + (-t)|_{0.1}^{1} \\ &= (0.1 - 0) + (-1 + 0.1) \\ &= -0.8 \end{aligned}$$

$$
\begin{aligned}
a_n &= \frac{2}{1}(\int_{-0.9}^{0} -cos(2\pi nt)\ dt + \int_{0}^{0.1} cos(2\pi nt)\ dt) \\
&= 2((-\frac{1}{2\pi n}sin(2\pi nt))|_{-0.9}^{0} + (\frac{1}{2\pi n}sin(2\pi nt))|_{0}^{0.1}) \\
&= 2((0 + \frac{sin(-1.8\pi n)}{2\pi n}) + (\frac{sin(0.2\pi n)}{2\pi n}) - 0)) \\
&= \frac{sin(-1.8\pi n) + sin(0.2\pi n)}{\pi n}
\end{aligned}
$$

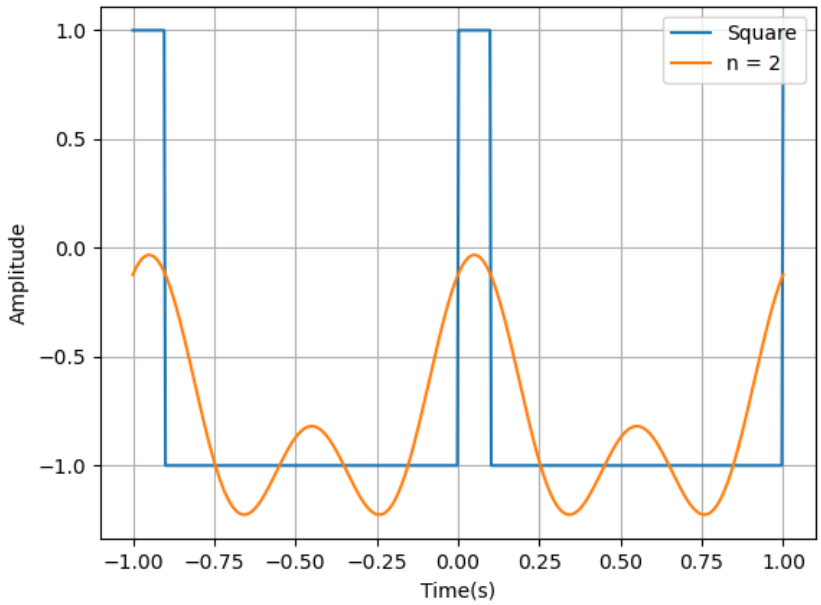
$$
\begin{aligned}
b_n &= \frac{2}{1}(\int_{-0.9}^{0} -sin(2\pi nt)\ dt + \int_{0}^{0.1} sin(2\pi nt)\ dt) \\
&= 2((\frac{cos(2\pi nt)}{2\pi n})|_{-0.9}^{0} + (-\frac{cos(0.2\pi n)}{2\pi n})|_{0}^{0.1}) \\
&= 2((\frac{1}{2\pi n} - \frac{cos(-1.8\pi n)}{2\pi n}) + (\frac{-cos(0.2\pi n)}{2\pi n} + \frac{1}{2\pi n})) \\
&= \frac{2}{\pi n} - \frac{cos(-1.8\pi n) + cos(0.2\pi n)}{\pi n} \\
&= \frac{2 - cos(-1.8\pi n) - cos(0.2\pi n)}{\pi n}
\end{aligned}
$$

$$
x(t) = -0.8 + \sum_{n=1}^{\infty}(\frac{sin(-1.8\pi n) + sin(0.2\pi n)}{\pi n}cos(2\pi nt) + \frac{2 - cos(-1.8\pi n) - cos(0.2\pi n)}{\pi n}sin(2\pi nt))
$$

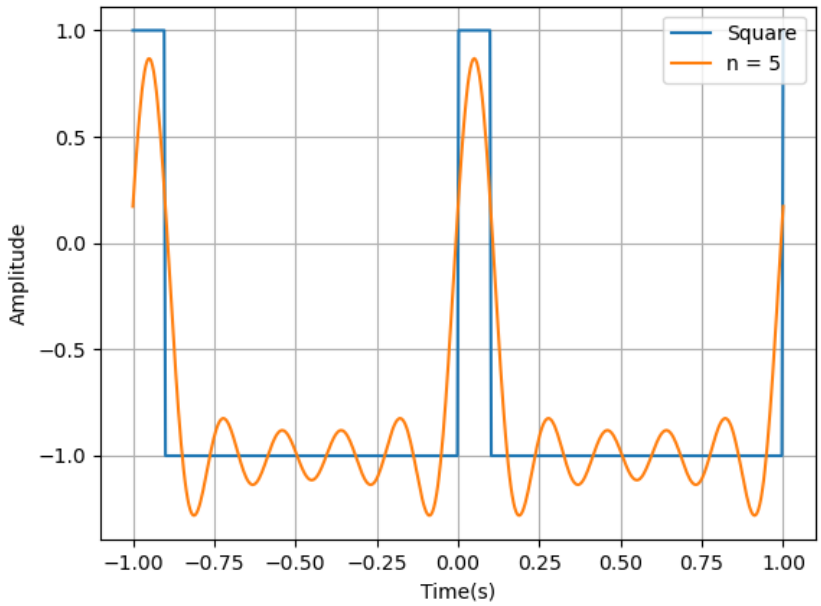
Question 3.1



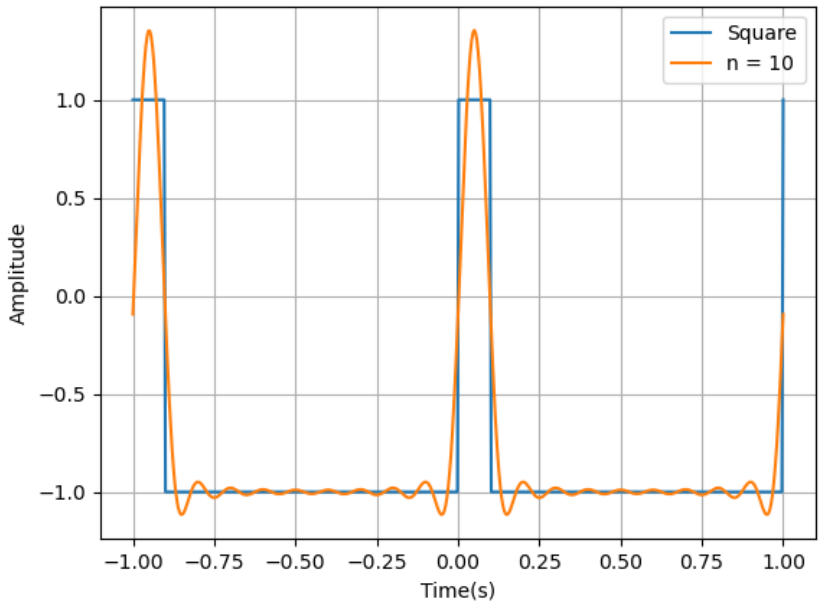
Question 3.2



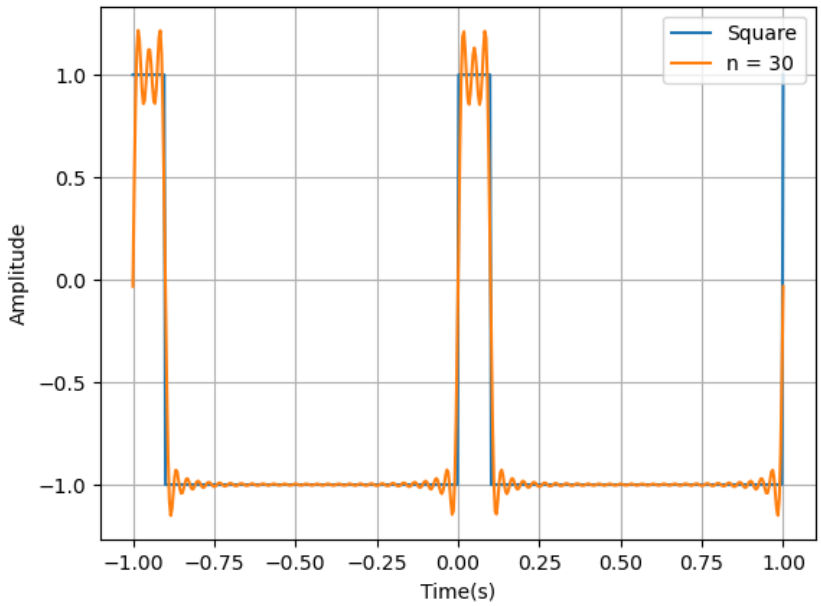
Question 3.3



Question 3.4



Question 3.5



$$x(t) = -0.8 + \sum_{n=1}^{\infty} \left(\frac{sin(-1.8\pi n) + sin(0.2\pi n)}{\pi n} cos(2\pi nt) + \frac{2 - cos(-1.8\pi n) - cos(0.2\pi n)}{\pi n} sin(2\pi nt) \right)$$

With n form 1 to 30 in the above Fourier series we will get a good approximation of a square wave with 10% duty.

Thus we can determine thirty components are needed to form a good graphical representation of a square wave with 10% duty cycle.

2.4 Question 4

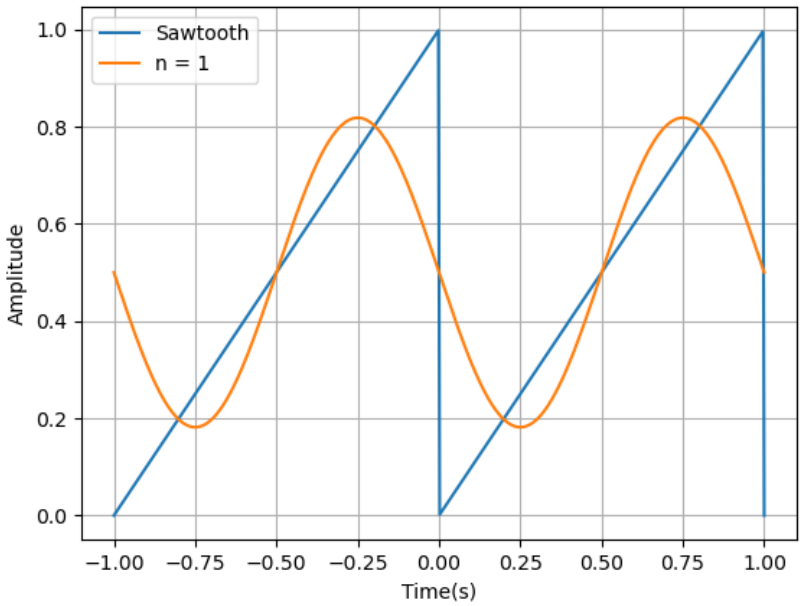
$$T = 0$$
$$\omega_0 = 2\pi$$
$$a_0 = 0$$
$$a_n = 0$$
$$0 \leq t \leq 1$$

$$
\begin{aligned}
D_n &= \frac{1}{1}\int_0^1 t.e^{-jn2\pi t}\ dt \\
&= (t\frac{e^{-jn2\pi t}}{-jn2\pi})|_0^1 - (\frac{1}{-jn2\pi})\int_0^1 e^{-jn2\pi t}\ dt \\
&= (t\frac{e^{-jn2\pi t}}{-jn2\pi})|_0^1 - (\frac{1}{-jn2\pi})(\frac{e^{-jn2\pi t}}{-jn2\pi})|_0^1 \\
&= (t\frac{e^{-jn2\pi t}}{-jn2\pi})|_0^1 - (\frac{e^{-jn2\pi t}}{-j^2(n2\pi)^2})|_0^1 \\
&= (t\frac{e^{-jn2\pi t}}{-jn2\pi})|_0^1 + (\frac{e^{-jn2\pi t}}{(n2\pi)^2})|_0^1 \\
&= (\frac{e^{-jn2\pi t}}{-jn2\pi} - 0) + (\frac{e^{-jn2\pi t}}{(n2\pi)^2} - \frac{1}{(n2\pi)^2}) \\
&= (\frac{1}{-jn2\pi}) + (\frac{1}{(n2\pi)^2} - \frac{1}{(n2\pi)^2}) \quad \text{(see reason bellow)} \\
D_n &= \frac{1}{-jn2\pi}
\end{aligned}
$$

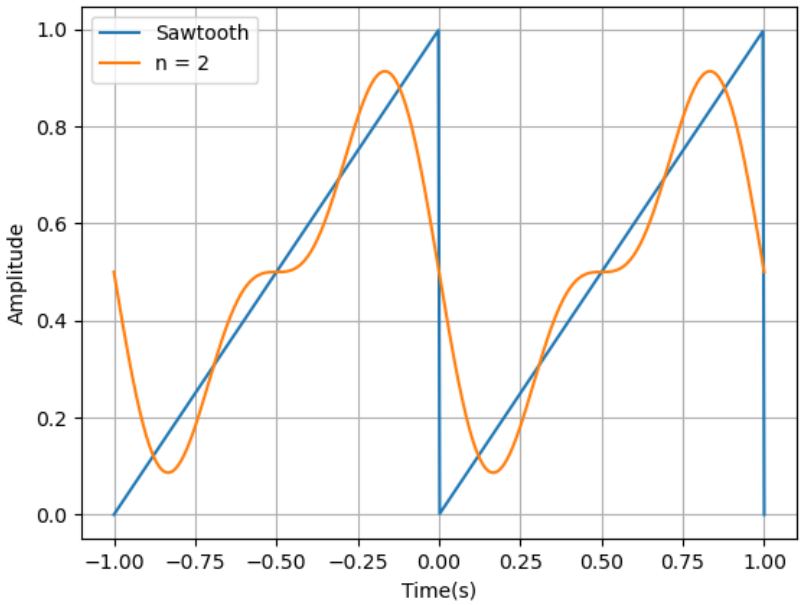
$$
\begin{gathered}
e^{-jn2\pi} = cos(n2\pi) - jsin(2n\pi) \\
= 1 - 0 \text{ for any interger value of n} \\
\text{thus, } e^{-jn2\pi} \to 1
\end{gathered}
$$

$$
f(t) = \sum_{n=-\infty}^{\infty} (\frac{1}{-jn2\pi})e^{jn2\pi t}
$$

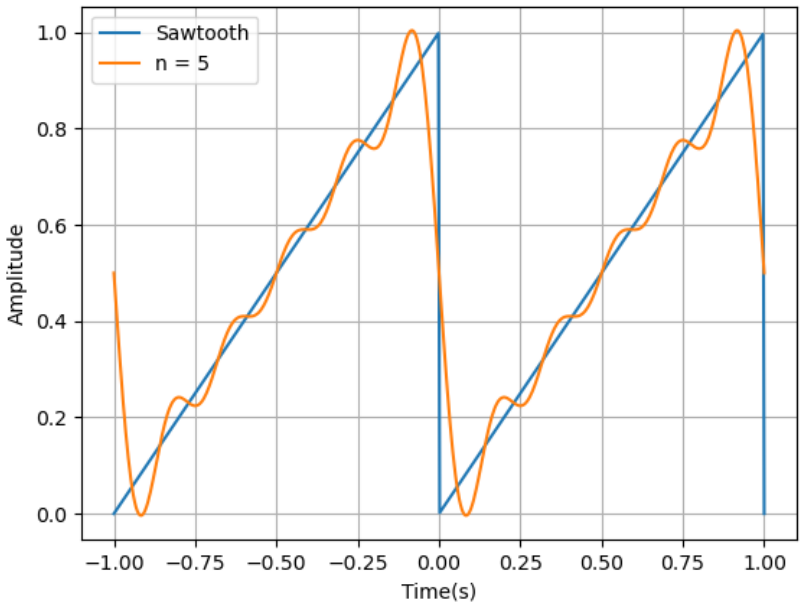
Question 4.1



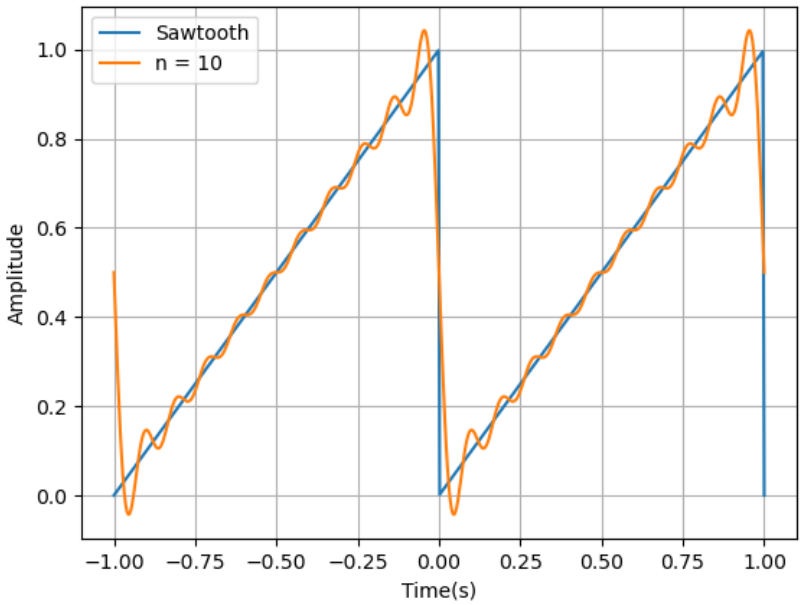
Question 4.2



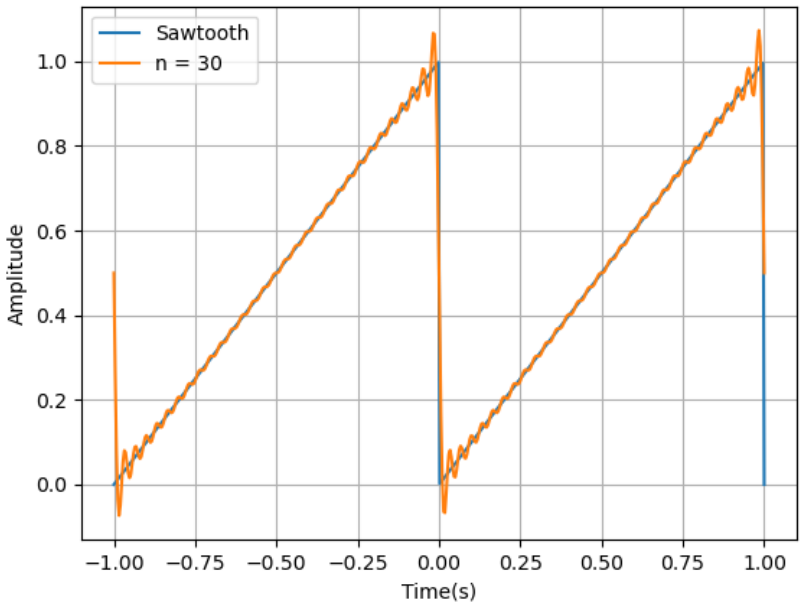
Question 4.3



Question 4.4



Question 4.5



$$f(t) = \sum_{n=-\infty}^{\infty} (\frac{1}{-jn2\pi})e^{jn2\pi t}$$

With n form 1 to 30 in the above Fourier series we will get a good approximation of a saw-tooth wave with function $x(t) = t, 0 \leq t \leq 1$.

Thus we can determine thirty components are needed to form a good graphical representation of a saw-tooth wave with function $x(t) = t, 0 \leq t \leq 1$.

3 Conclusion

Through the use of appropriate calculations and the Python programming language, we were able to effectively approximate and plot the Fourier expansions of several periodic signals. We were also able to determine how many components of the Fourier series were needed to effectively plot an accurate graph.

Thus it was a successful experiment as we were able to achieve all expected practical outcomes.

4 References

[1] B. Lathi, "Continious time signal analysis Fourier series," in Linear systems and signal, New York, Oxford university press, 2010, pp. 594-640.

[2] "Stack Overflow," November 2015. [Online]. Available: https://stackoverflow.com/questions/32669101/how-to-define-periodic-function-in-python. [Accessed 8 September 2020].

5 Appendix A: Code

Question 1:

```
import matplotlib.pyplot as plt
import numpy as np
from scipy import signal

t = np.linspace(-1, 1, 500, endpoint=True)
plt.plot(t, np.sin(2*np.pi*t), label="Sine")
plt.xlabel("Time(s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.legend()
plt.show()
```

Question 2:

```
import matplotlib.pyplot as plt
import numpy as np
from scipy import signal

def cn(n):
    return (2*(np.cos(np.pi*n)-1))/(np.pi*n)

def xT(n_max, x):
    xt = 0.0
    for n in range(1, n_max+1):
        xt += (cn(n)*np.cos(2*np.pi*n*x + np.pi/2))
    return xt


t = np.linspace(-1, 1, 500, endpoint=True)
y = np.array([xT(1, i) for i in t])
y2 = np.array([xT(2, i) for i in t])
y3 = np.array([xT(5, i) for i in t])
y4 = np.array([xT(10, i) for i in t])
y5 = np.array([xT(40, i) for i in t])
plt.plot(t, signal.square(2*np.pi*t, 0.5), label="Square")
plt.plot(t, y, label="n=1")
plt.plot(t, y2, label="n=2")
plt.plot(t, y3, label="n=5")
plt.plot(t, y4, label="n=10")
plt.plot(t, y5, label="n=40")
plt.xlabel("Time(s)")
plt.ylabel("Amplitude")
```

```
plt.grid(True)
plt.legend()
plt.show()
```

Question 3:

```
import matplotlib.pyplot as plt
import numpy as np
from scipy import signal

def an(n):
    return (np.sin(-1.8*np.pi*n)+np.sin(0.2*np.pi*n))/(np.pi*n)

def bn(n):
    return (2-np.cos(-1.8*np.pi*n)-np.cos(0.2*np.pi*n))/(np.pi*n)

def xT(n_max, x):
    xt = -0.8
    for n in range(1, n_max+1):
        xt += ((an(n)*np.cos(2*np.pi*n*x))+(bn(n)*np.sin(2*np.pi*n*x)))
    return xt


t = np.linspace(-1, 1, 500, endpoint=True)
y = np.array([xT(1, i) for i in t])
y2 = np.array([xT(2, i) for i in t])
y3 = np.array([xT(5, i) for i in t])
y4 = np.array([xT(10, i) for i in t])
plt.plot(t, signal.square(2*np.pi*t, 0.1), label="Square")
plt.plot(t, y, label="n = 1")
plt.plot(t, y2, label="n = 2")
plt.plot(t, y3, label="n = 5")
plt.plot(t, y4, label="n = 10")
plt.xlabel("Time(s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.legend()
plt.show()
```

Question 4:

```
import matplotlib.pyplot as plt
import numpy as np

def periodicSawtooth(a, b):
```

```
    interval = b - a
    return lambda f: lambda x: f((x - a) % interval + a)

@periodicSawtooth(0, 1)
def f(x):
    return x

def dn(n):
    return 2*(-2*np.exp(-2j*np.pi*n)*np.pi*n + 1j*(np.exp(-2j*np.pi*n)-1))
    /(4j*pow(np.pi, 2)*pow(n, 2))

def xT(n_max, x):
    xt = 0.5
    for n in range(1, n_max+1):
        xt += (dn(n))*(np.exp(1j*2*np.pi*n*x))
    return xt

t = np.linspace(-1, 1, 500, endpoint=True)
sawtooth = periodicSawtooth(0, 1)(lambda x: x)
y = np.array([xT(1, i) for i in t])
y2 = np.array([xT(2, i) for i in t])
y3 = np.array([xT(5, i) for i in t])
y4 = np.array([xT(10, i) for i in t])
plt.plot(t, sawtooth(t), label="Sawtooth")
plt.plot(t, y, label="n = 1")
plt.plot(t, y2, label="n = 2")
plt.plot(t, y3, label="n = 5")
plt.plot(t, y4, label="n = 10")
plt.xlabel("Time(s)")
plt.ylabel("Amplitude")
plt.grid(True)
plt.legend()
plt.show()
```